Quantum-Optimized Spatial Permutations: Minimizing Submatrix Rank for Block-Low-Rank Space-Filling Curves

Amr Magdy
University of California, Riverside
Riverside, California, USA
amr@cs.ucr.edu

Yunhan Chang
University of California, Riverside
Riverside, California, USA
ychan268@ucr.edu

Sameh Abdulah
King Abdullah University of Science and Technology
Thuwal, Saudi Arabia
sameh.abdulah@kaust.edu.sa

Abstract

We propose a fundamental extend from classical, geometrically recursive space-filling curves (SFCs). Rather than preserving Euclidean proximity, we frame spatial layout generation as a quantum-native combinatorial optimization problem that focuses on statistical correlations. Given an $N \times N$ spatial covariance matrix and a tile size n_b, we seek a permutation that improves the block structure of the permuted covariance matrix for tile low-rank (TLR) approximation. In standard TLR methods, diagonal tiles are typically retained in dense form, while off-diagonal tiles are compressed using low-rank representations. Therefore, the goal is not to minimize the rank of the diagonal tiles themselves. Instead, we aim to construct a permutation that groups highly correlated locations into common diagonal tiles, thereby reducing cross-tile covariance mass and potentially improving the numerical compressibility of the off-diagonal TLR blocks. Because exact rank minimization is mathematically intractable and non-differentiable, we map this objective to a Constrained Quadratic Model (CQM) executable on modern quantum annealers. By utilizing a quadratic surrogate objective that maximizes the Frobenius norm within the diagonal blocks, we concentrate data variance to achieve block-low-rank structures. This approach enables highly correlated, maximally compressible spatial data layouts that bridge advanced linear algebra, quantum optimization, and foundational spatial database theory.

ACM Reference Format:
Amr Magdy, Yunhan Chang, and Sameh Abdulah. 2026. Quantum-Optimized Spatial Permutations: Minimizing Submatrix Rank for Block-Low-Rank Space-Filling Curves. In *The third workshop on Quantum Computing and Quantum-Inspired Technology for Data-Intensive Systems and Applications (Q-Data '26), May 31–June 05, 2026, Bengaluru, India.* ACM, New York, NY, USA, 2 pages. https://doi.org/10.1145/3811628.3811833

1 Background

Gaussian process models capture spatial dependence but are limited by dense covariance costs, which TLR approximation reduces by exploiting matrix structure. We briefly review both in this section.

Q-Data '26, Bengaluru, India

ACM ISBN 979-8-4007-2703-0/26/05
https://doi.org/10.1145/3811628.3811833

1.1 Spatial Statistics and Gaussian Processes

Spatial statistics provides a principled framework for modeling data observed over geographical locations, where dependence between observations is governed by spatial proximity [6]. A common approach is to model such data as a realization of a Gaussian process (GP), where observations $\mathbf{Z} = \{Z(s_1), \ldots, Z(s_n)\}^\top$ follow a multivariate normal distribution with mean and covariance defined through a parametric function such as the Matérn class. The resulting dense covariance matrix $\Sigma(\boldsymbol{\theta})$ encodes pairwise spatial correlations for inference and prediction.

Parameter estimation is typically performed via maximum likelihood estimation (MLE), which relies on the Gaussian log-likelihood function [1]:

$$\ell(\boldsymbol{\theta}) = -\frac{n}{2}\log(2\pi) - \frac{1}{2}\log|\Sigma(\boldsymbol{\theta})| - \frac{1}{2}\mathbf{Z}^\top\Sigma(\boldsymbol{\theta})^{-1}\mathbf{Z}. \quad (1)$$

Evaluating it requires log-determinants and dense linear solves, typically via Cholesky factorization. This costs $O(n^3)$ time and $O(n^2)$ memory, becoming prohibitive at large spatial scales. Consequently, scalable approximation techniques are essential to enable practical Gaussian process modeling in modern high-resolution applications.

1.2 Tile Low-Rank Approximation

The computational bottleneck in Gaussian process modeling arises directly from the need to operate on large, dense covariance matrices. Although these matrices are formally dense, they often exhibit a data-sparse structure because spatial correlations decay with distance. This implies that off-diagonal blocks of the covariance matrix can be well-approximated by low-rank representations. Tile Low-Rank (TLR) approximation exploits this property to significantly reduce computational and memory costs [2, 8]. In the TLR framework, the covariance matrix is partitioned into smaller square tiles (individual blocks). While diagonal tiles are retained in dense form, each off-diagonal tile can be approximated using a truncated singular value decomposition (SVD), yielding a factorized representation $\mathbf{U}_{ij}\mathbf{V}_{ij}^\top$ with rank $k \ll n_b$, where n_b is the tile size. This compression reduces both storage requirements and arithmetic complexity, enabling efficient computation of key operations such as Cholesky factorization and likelihood evaluation on the compressed structure. As a result, TLR-based methods can substantially reduce the effective complexity relative to the cubic scaling of dense approaches.

Beyond computational savings, TLR is particularly well-suited for modern high-performance computing (HPC) architectures. Its tile-based structure enables fine-grained parallelism, allowing independent tasks on tiles to be scheduled across CPUs and GPUs, thereby improving scalability and resource utilization. The effectiveness of TLR depends on factors such as tile size, approximation accuracy, and spatial ordering of locations, all of which influence the rank of off-diagonal tiles and ultimately determine the trade-off between accuracy and performance. In particular, the ordering of spatial locations prior to constructing the covariance matrix is critical for maximizing compression efficiency [5]. When locations that are close in space are also placed close in the matrix ordering, the resulting covariance matrix exhibits a stronger block structure, with large values concentrated near the diagonal and rapidly decaying off-diagonal interactions. This structure leads to lower numerical ranks in the off-diagonal tiles, enabling more aggressive compression and motivating the covariance-aware reordering formulation introduced next.

2 Methodology and Theoretical Formulation

Classical space-filling curves, such as the Hilbert or Z-curves, rely on deterministic, recursive fractals to map multi-dimensional spatial data into a one-dimensional sequence [3]. While mathematically elegant, these geometric heuristics fail to capture the complex, data-driven statistical correlations present in real-world spatial distributions. They rigidly enforce Euclidean locality but do not optimize for the actual information entropy or compressibility of the data as it is loaded into memory blocks. We propose abandoning these heuristic geometries entirely. Instead, we redefine the generation of a spatial layout as an objective-driven permutation problem, optimized directly for block-based spatial compression and tensor operations.

Our optimization goal departs from simple pairwise distance preservation. Let $\Sigma \in \mathbb{R}^{N\times N}$ be the dense spatial covariance matrix capturing the statistical relationships between N distinct spatial locations. In a block-based database system with a memory tile capacity of T locations, data is loaded in chunks. To maximize compression and computational efficiency, the data within each tile must be maximally correlated. Mathematically, we seek a permutation matrix $P \in \{0,1\}^{N\times N}$ that reorders the locations to form a permuted covariance matrix $\Sigma' = P\Sigma P^T$. We then partition Σ' into disjoint $T \times T$ submatrices along its main diagonal. The desired permutation groups strongly correlated locations into common diagonal tiles, so that cross-tile covariance mass is reduced.

Because computing the true rank of a matrix is a discrete, nonlinear, and non-differentiable operation, it cannot be directly formulated as an objective function for quantum annealers. To execute this on quantum hardware, we must map the problem to a Constrained Quadratic Model (CQM)[7] using a continuous surrogate objective. Since the total Frobenius norm of Σ is invariant under simultaneous row-column permutation, we use the squared covariance mass within same-tile groups as a tractable surrogate for covariance concentration. This objective does not directly minimize matrix rank; instead, it encourages highly correlated locations to be assigned to common tiles. The resulting permutation should therefore be evaluated through downstream TLR metrics, such as off-diagonal numerical ranks, compression ratios, factorization time, and likelihood accuracy. Concentrating covariance mass within same-tile groups does not change the values of covariance entries; it only changes their positions in the permuted matrix. The intended effect is to place large covariance entries inside tiles, leaving weaker covariance interactions across different tiles and potentially improving the compressibility of off-diagonal TLR blocks.

We define a binary decision variable $x_{i,k} \in \{0,1\}$, where $x_{i,k} = 1$ if spatial location $i \in \{1,\ldots,N\}$ is assigned to sequence position $k \in \{1,\ldots,N\}$. To identify which sequence positions belong to the same memory tile, we define an indicator function $\delta_T(k,m) \in \{0,1\}$, which equals 1 if $\lfloor (k-1)/T \rfloor = \lfloor (m-1)/T \rfloor$, and 0 otherwise. We formulate the surrogate quadratic objective function as the maximization of the intra-block variance:

$$\max \sum_{i=1}^{N}\sum_{j=1}^{N}\sum_{k=1}^{N}\sum_{m=1}^{N} \delta_T(k,m)\Sigma_{i,j}^2 x_{i,k}x_{j,m}$$

To ensure that the resulting matrix P is a valid permutation matrix (i.e., a valid space-filling curve without duplications or omissions), we enforce two strict, one-hot linear constraints natively supported by the CQM architecture. First, every spatial location i must be assigned to exactly one sequence position: $\sum_{k=1}^{N} x_{i,k} = 1$ for all i. Second, every sequence position k must contain exactly one spatial location: $\sum_{i=1}^{N} x_{i,k} = 1$ for all k.

By expressing this formulation as a Constrained Quadratic Model (CQM), the permutation search becomes accessible to hybrid quantum-classical annealing solvers, including D-Wave's CQM solver. This does not imply a guaranteed quantum speedup; rather, it provides an alternative heuristic search framework for comparing covariance-aware layouts against classical spatial orderings [4]. This quantum-native optimization synthesizes an entirely new class of space-filling curves, curves governed not by fractal geometry, but by the intrinsic covariance and low-rank structural potential of the spatial data itself.

References

[1] Sameh Abdulah, Hatem Ltaief, Ying Sun, Marc G Genton, and David E Keyes. 2018. ExaGeoStat: A high performance unified software for geostatistics on manycore systems. *IEEE Transactions on Parallel and Distributed Systems* 29, 12 (2018), 2771–2784.

[2] Kadir Akbudak, Hatem Ltaief, Aleksandr Mikhalev, and David Keyes. 2017. Tile low rank Cholesky factorization for climate/weather modeling applications on manycore architectures. In *International Conference on High Performance Computing*. Springer, 22–40.

[3] Yunhan Chang, Amr Magdy, and Federico Spedalieri. 2025. Quantum Modeling of Spatial Contiguity Constraints *(Q-Data '25)*. Association for Computing Machinery, New York, NY, USA, 3–9. doi:10.1145/3736393.3736691

[4] Yunhan Chang, Amr Magdy, Federico Spedalieri, and Ibrahim Sabek. 2025. Spatial Regionalization: A Hybrid Quantum Computing Approach. In *Proceedings of the 33rd ACM International Conference on Advances in Geographic Information Systems* (The Graduate Hotel Minneapolis, Minneapolis, MN, USA) *(SIGSPATIAL '25)*. Association for Computing Machinery, New York, NY, USA, 1130–1133. doi:10.1145/3748636.3763211

[5] Sihan Chen, Sameh Abdulah, Ying Sun, and Marc G Genton. 2024. On the impact of spatial covariance matrix ordering on tile low-rank estimation of Matérn parameters. *Environmetrics* 35, 6 (2024), e2868.

[6] Noel Cressie. 2015. *Statistics for spatial data*. John Wiley & Sons.

[7] Catherine McGeoch et al. 2022. Constrained Quadratic Models for Quantum Annealing. *Quantum Tech Repts* (2022).

[8] Sagnik Mondal, Sameh Abdulah, Hatem Ltaief, Ying Sun, Marc G Genton, and David E Keyes. 2023. Tile low-rank approximations of non-Gaussian space and space-time Tukey g-and-h random field likelihoods and predictions on large-scale systems. *J. Parallel and Distrib. Comput.* 180 (2023), 104715.